MyWay Tech Report: Map Providers & GPX Libraries

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1 GPX Libraries

- **JPX**: (*Java*, already in use) `https://github.com/jenetics/jpx` - Primarily utilized for backend GPX parsing and validation. Key advantages include strict immutability, thread safety, and built-in spatial computations.
- **Leaflet**: (*JavaScript/TypeScript*, already in use) `https://leafletjs.com` - Mostly applied to render interactive 2D maps. Major benefits involve a tiny bundle size and excellent cross-platform DOM performance.
- **MapLibre GL JS / Mapbox GL JS**: (*JavaScript/TypeScript*) `https://maplibre.org` / `https://docs.mapbox.com/mapbox-gl-js` - Chiefly deployed for high-performance WebGL client-side rendering. Notable perks are hardware acceleration, fluid 3D terrain elevation, and the ability to handle massive datasets effortlessly.
- **OpenLayers**: (*JavaScript*) `https://openlayers.org` - Typically chosen for complex enterprise GIS web applications. Standout features encompass native GPX parsing without external plugins and robust spatial projection support.
- **@mapbox/togeojson**: (*JavaScript/Node.js*) `https://github.com/mapbox/togeojson` - Principally serves to convert GPX and KML datasets into GeoJSON format. Its primary strengths lie in rapid execution and seamless integration with modern WebGL pipelines.

2 Map Providers

- **OpenStreetMap (OSM)**: (*Raster Map Tiles*, already in use) `https://www.openstreetmap.org` - Predominantly relied upon for global topographic and road network displays. The greatest benefits are its completely free, community-driven nature, and suitability for open-source outdoor apps.

- **Mapbox / MapLibre**: (*Vector Tiles*) `https://www.mapbox.com` - Generally employed for advanced custom map visualizations. Top advantages consist of high-performance rendering, dynamic styling capabilities, and native 3D terrain support.
- **Thunderforest / MapTiler**: (*Specialized Map Tiles*) `https://www.thunderforest.com` / `https://www.maptiler.com` - Mostly targeted at dedicated hiking and trail applications. Winning traits include high-resolution outdoor layers and highly detailed topographic datasets.